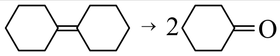
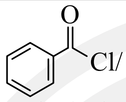
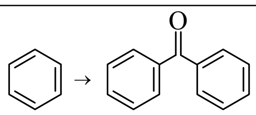
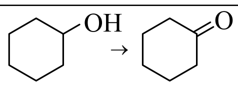
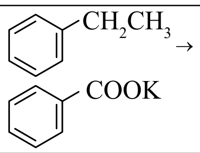
Chapter 9

Aldehydes, Ketones and Carboxylic Acids

Preparation of Aldehydes and Ketones

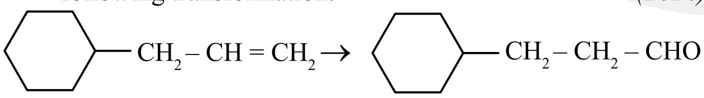
1. Match List I with List II. *(2024)*

	List I (Reaction)		List II (Reagents/Condition)
A.		I.	 Anhyd.$AlCl_3$
B.		II.	CrO_3
C.		III.	$KMnO_4$/KOH, Δ
D.		IV.	(i) O_3 (ii) Zn-H_2O

Choose the correct answer from the options given below:

a. A-IV, B-I, C-II, D-III b. A-I, B-IV, C-II, D-III
c. A-IV, B-I, C-III, D-II d. A-III, B-I, C-II, D-IV

2. Identify the correct reagents that would bring about the following transformation. *(2024)*



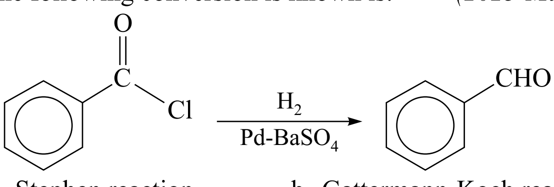
a. (i) BH_3
(ii) $H_2O_2/\overset{\ominus}{O}H$
(iii) alk. $KMnO_4$
(iv) $H_3O^{\oplus}$

b. (i) H_2O/H^+
(ii) PCC

c. (i) H_2O/H^+
(ii) CrO_3

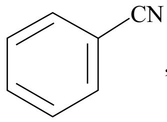
d. (i) BH_3
(ii) $H_2O_2/\overset{\ominus}{O}H$
(iii) PCC

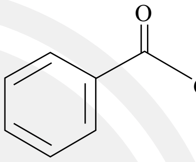
3. The following conversion is known is: *(2023-Manipur)*

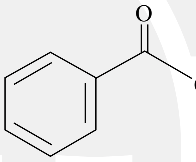


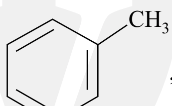
a. Stephen reaction b. Gattermann-Koch reaction
c. Etard reaction d. Rosenmund reaction

4. The incorrect method to synthesize benzaldehyde is: *(2022 Re)*

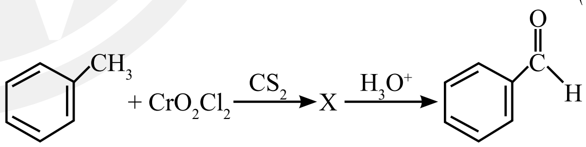
a.  , CH_3MgBr, followed by H_3O^+

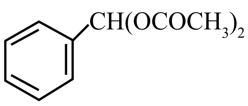
b.  Cl, H_2, Pd-$BaSO_4$

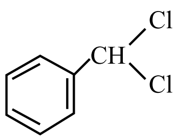
c.  OC_2H_5, DIBAL-H followed by H_2O

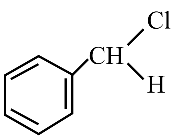
d.  , CrO_2Cl_2, followed by H_3O^+ in CS_2

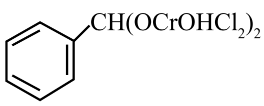
5. The intermediate compound 'X' in the following chemical reaction is: *(2021)*



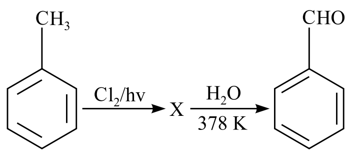
a. 

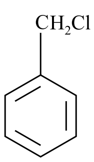
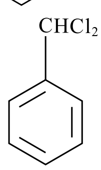
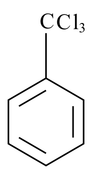
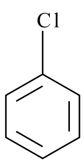
b. 

c. 

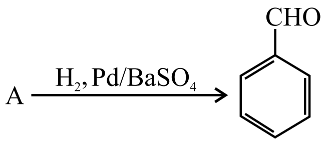
d. 

6. Identify compound X in the following sequence of reactions: *(2020)*



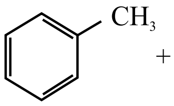
a.  b.  c.  d. 

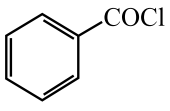
7. Identify compound (A) in the following reaction: *(2020-Covid)*

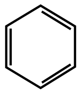


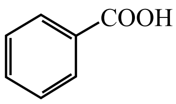
a. Toluene
b. Acetophenone
c. Benzoic acid
d. Benzoyl chloride

8. Reaction by which Benzaldehyde cannot be prepared: *(2013)*

a.  + CrO_2Cl_2 in CS_2 followed by H_3O^+

b.  + H_2 in presence of Pd-$BaSO_4$

c.  + CO + HCl in presence of anhydrous + $AlCl_3$

d.  + Zn/Hg and concentration HCl

Physical Properties and Chemical Reactions of Aldehydes and Ketones

9. Identify D in the following sequence of reactions: *(2024 Re)*

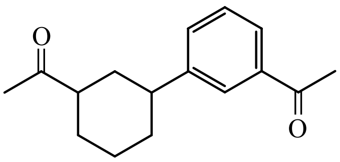
$$CH_3CH_2OH \xrightarrow{P+I_2} A \xrightarrow[\text{dry ether}]{Mg} B \xrightarrow{HCHO} C \xrightarrow{H_2O} \underset{\text{(major)}}{D}$$

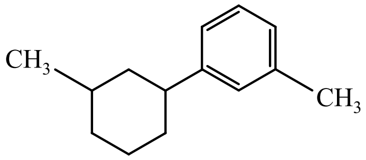
a. n-propyl alcohol
b. isopropyl alcohol
c. propanal
d. propionic acid

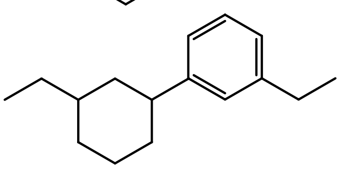
10. Fehling's solution 'A' is *(2024)*

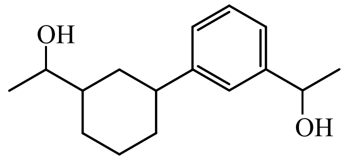
a. alkaline solution of sodium potassium tartrate (Rochelle's salt)
b. aqueous sodium citrate
c. aqueous copper sulphate
d. alkaline copper sulphate

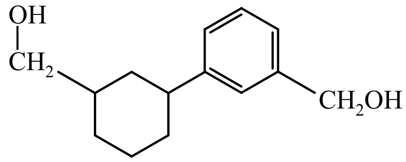
11. Identify product (A) in the following reaction: *(2023)*

 $\xrightarrow[\text{Conc.HCl}]{\text{Zn–Hg}}$ (A) + $2H_2O$

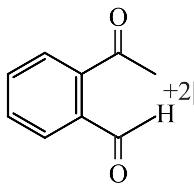
a. 

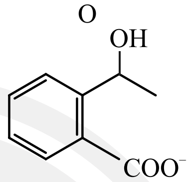
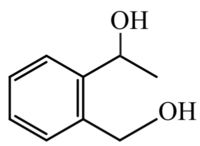
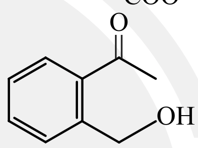
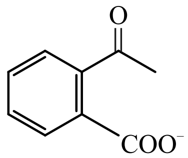
b. 

c. 

d. 

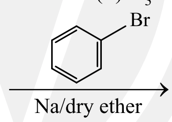
12. Identify the major product obtained in the following reaction: *(2023)*

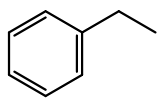
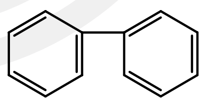
 $+2[Ag(NH_3)_2]^+ + 3\ OH^- \xrightarrow{\Delta}$ major product

a. 
b. 
c. 
d. 

13. Identify the final product [D] obtained in the following sequence of reactions, *(2023)*

$$CH_3CHO \xrightarrow[\text{(ii) } H_3O^+]{\text{(i) } LiAlH_4} [A] \xrightarrow[\Delta]{H_2SO_4} [B] \xrightarrow{HBr}$$

[C]  [D]

a. $HC \equiv C^-Na^+$
b. 
c. 
d. C_4H_{10}

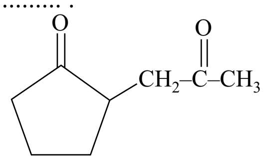
14. Consider the given reaction: *(2023-Manipur)*

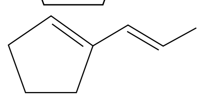
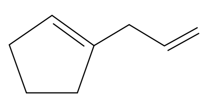
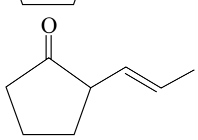
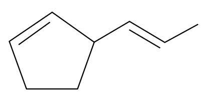
$$CH_3COCH_3 \xrightarrow{\text{dil } Ba(OH)_2} \text{“X”}$$

The functional groups present in compound "X" are:

a. ketone are double bond
b. double bond and aldehyde
c. Alcohol and aldehyde
d. alcohol and ketone

15. The **major** product formed in the following conversion is .... *(2023-Manipur)*

 $\xrightarrow[\text{(ii) } H_2SO_4,\ \Delta]{\text{(i) } NaBH_4}$ Major product

a. 
b. 
c. 
d. 

16. Which of the following reactions is not an example for nucleophilic addition – elimination reaction ? *(2022 Re)*

a. $CH_3CHO + NH_3 \rightleftharpoons CH_3CH = NH + H_2O$

b. $CH_3CHO + NaHSO_3 \rightleftharpoons CH_3-\underset{H}{\overset{OH}{C}}-OSO_2Na$

c. $CH_3CHO + NH_2OH \rightleftharpoons CH_3CH = N - OH + H_2O$

d. $CH_3CHO + C_6H_5NHNH_2 \rightleftharpoons CH_3CH = N–NHC_6H_5 + H_2O$

17. Given below are two statements: *(2022)*

Statement-I: The boiling points of aldehydes and ketones are higher than hydrocarbons of comparable molecular mass because of weak molecular association in aldehydes and ketones due to dipole - dipole interactions.

Statement-II : The boiling points of aldehydes and ketones are lower than the alcohols of similar molecular mass due to the absence of H-bonding.

In the light of the above statements, choose the **most appropriate** answer from the options given below

a. **Statement-I** is incorrect but **Statement-II** is correct.
b. Both **Statement-I** and **Statement-II** are correct.
c. Both **Statement-I** and **Statement-II** are incorrect.
d. **Statement-I** is correct but **Statement-II** is incorrect.

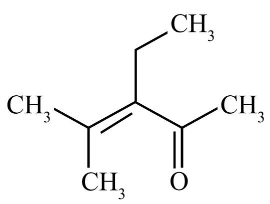
18. Match List-I with List-II. *(2022)*

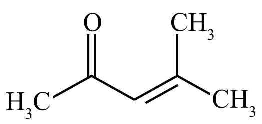
List-I (Products formed)		List-II (Reaction of carbonyl compound with)	
(a)	Cyanohydrin	(i)	NH_2OH
(b)	Acetal	(ii)	RNH_2
(c)	Schiff's base	(iii)	alcohol
(d)	Oxime	(iv)	HCN

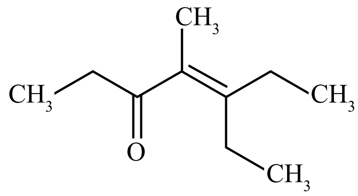
Choose the correct answer from the options given below:

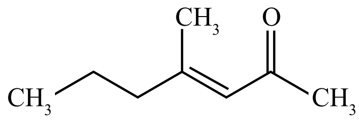
a. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
b. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
c. (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
d. (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

19. Which one of the following is not formed when acetone reacts with 2-pentanone in the presence of dilute NaOH followed by heating? *(2022)*

a. 

b. 

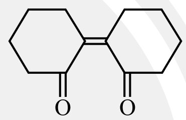
c. 

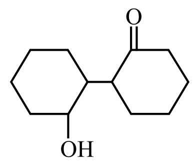
d. 

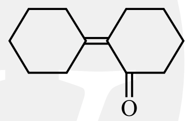
20. Reaction between benzaldehyde and acetophenone in presence of dilute NaOH is known as: *(2020)*

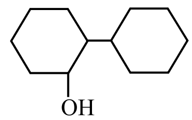
a. Cannizzaro's reaction
b. Cross Cannizzaro's reaction
c. Cross Aldol condensation
d. Aldol condensation

21. Which of the following product is formed when cyclohexanone undergoes aldol condensation followed by heating? *(2017-Delhi)*

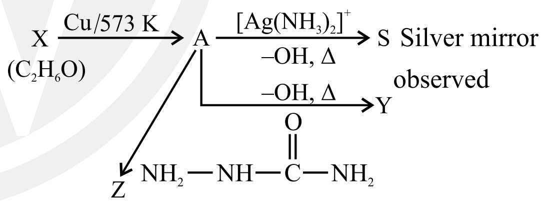
a. 

b. 

c. 

d. 

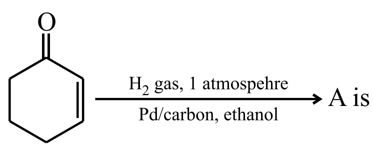
22. Consider the reactions: *(2017-Delhi)*

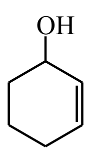


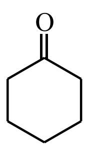
Identify A, X, Y and Z

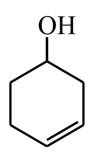
a. A-Ethanol, X-Acetaldehyde, Y-Butanone, Z-Hydrazone
b. A-Methoxymethane, X-Ethanoic acid, Y-Acetate ion, Z-hydrazine
c. A-Methoxymethane, X-Ethanol, Y-Ethanoic acid, Z-Semicarbazide
d. A-Ethanal, X-Ethanol, Y-But-2-enal, Z-Semicarbazone

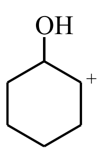
23. The correct structure of the product 'A' formed in the reaction *(2016 - II)*



a. 

b. 

c. 

d. 

24. The product formed by the reaction of an aldehyde with a primary amine is: *(2016 - I)*

a. Aromatic acid b. Schiff base
c. Ketone d. Carboxylic acid

25. Which of the following reagents would distinguish cis-cyclopenta-1, 2-diol from the trans-isomer? *(2016 - I)*

a. MnO_2 b. Aluminium isopropoxide
c. Acetone d. Ozone

26. The correct statement regarding a carbonyl compound with a hydrogen atom on its alpha-carbon, is: *(2016 - I)*

a. A carbonyl compound with a hydrogen atom on its alpha-carbon rapidly equilibrates with its corresponding enol and this process is known as carbonylation
b. A carbonyl compound with a hydrogen atom on its alpha-carbon rapidly equilibrates with its corresponding enol and this process is known as keto-enol tautomerism
c. A carbonyl compound with a hydrogen atom on its alpha-carbon never equilibrates with its corresponding enol
d. A carbonyl compound with a hydrogen atom on its alpha-carbon rapidly equilibrates with its corresponding enol and this process is known as aldehyde-ketone equilibration

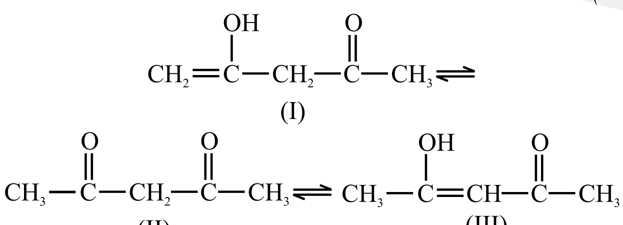
27. Reaction of a carbonyl compound with one of the following reagents involves nucleophilic addition followed by elimination of water. The reagent is: *(2015 Re)*

a. Sodium hydrogen sulphite
b. A Grignard reagent
c. Hydrazine in presence of feebly acidic solution
d. Hydrocyanic acid

28. An organic compound 'X' having molecular formula $C_5H_{10}O$ yields phenyl hydrazone and gives negative response to the Iodoform test and Tollen's test. It produces n-pentane on reduction. 'X' could be: *(2015)*

a. 2-pentanone b. 3-pentanone
c. n-pentyl alcohol d. Pentanal

29. The order of stability of the following tautomeric compounds is: *(2013)*

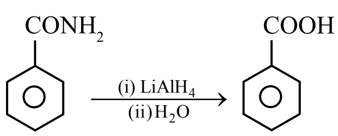


a. II > I > III b. II > III > I
c. I > II > III d. III > II > I

Methods of Preparation of Carboxylic Acids

30. Select the incorrect reaction among the following: *(2024 Re)*

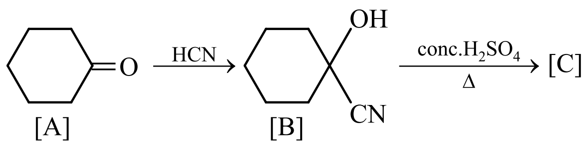
a. $CH_3COCl \xrightarrow[\Delta]{H_2O} CH_3COOH$

b. 

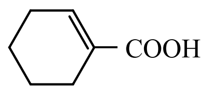
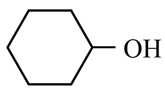
c. $CH_3CH_2OH \xrightarrow[\text{(ii)}H_3O^{\oplus}]{\text{(i)}KMnO_4/OH^-} CH_3COOH$

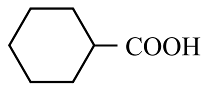
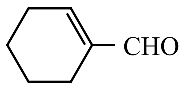
d. $CH_3CH_2CH_2OH \xrightarrow{CrO_3-H_2SO_4} CH_3CH_2COOH$

31. Complete the following reaction : *(2023)*



[C] is ________

a.  b. 

c.  d. 

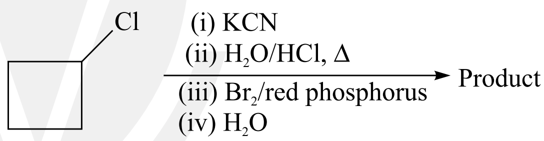
32. $RMgX + CO_2 \xrightarrow[\text{ether}]{\text{dry}} Y \xrightarrow{H_3O^+} RCOOH$ *(2022)*

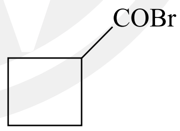
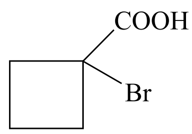
What is Y in the above reaction?

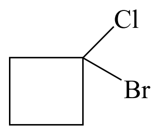
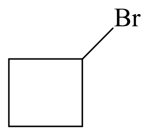
a. $(RCOO)_2Mg$ b. $RCOO^-Mg^+X$
c. $R_3CO^-Mg^+X$ d. $RCOO^-X^+$

Physical Properties and Chemical Reactions of Carboxylic Acids

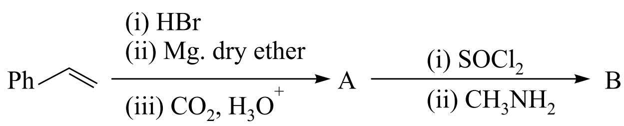
33. Identify the product in the following reaction *(2023-Manipur)*

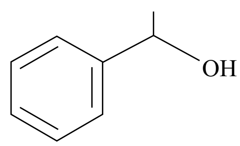


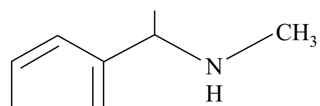
a.  b. 

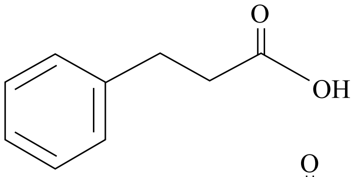
c.  d. 

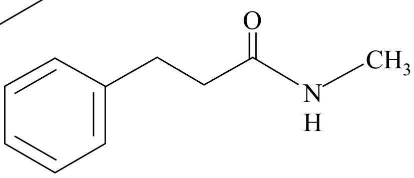
34. The products A and B in the following reaction sequence are: *(2022 Re)*

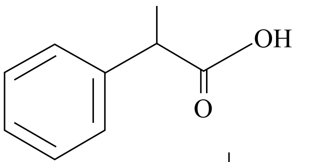


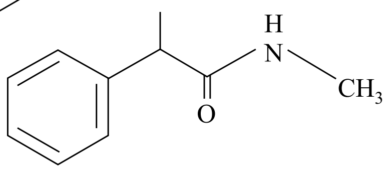
a. A = 

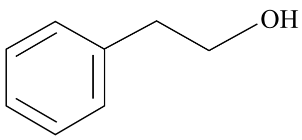
B = 

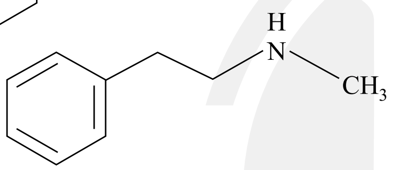
b. A = 

B = 

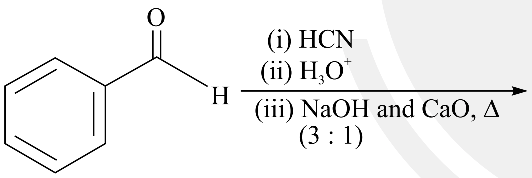
c. A = 

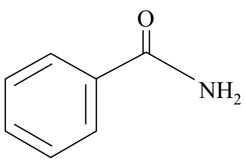
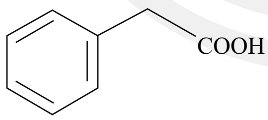
B = 

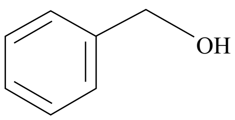
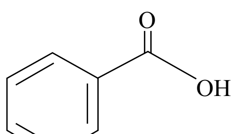
d. A = 

B = 

35. The product formed from the following reaction sequence is: *(2022 Re)*



a.  b. 

c.  d. 

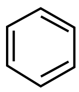
36. $CH_3CH_2COO^-Na^+ \xrightarrow[\text{Heat}]{\text{NaOH, +?}} CH_3CH_3 + Na_2CO_3$.

Consider the above reaction and identify the missing reagent/chemical. *(2021)*

a. Red Phosphorus b. CaO

c. DIBAL-H d. B_2H_6

37. Match List-I with List-II. *(2021)*

	List-I		List-II
(A)	 $\xrightarrow[\text{Anhyd. } AlCl_3/CuCl]{\text{CO, HCl}}$	(i)	Hell-Volhard-Zelinsky reaction
(B)	$R-\overset{\overset{O}{\|}}{C}-CH_3 +$ NaOX $\rightarrow$	(ii)	Gattermann-Koch reaction
(C)	$R-CH_2-OH + R'COOH \xrightarrow{\text{Conc. } H_2SO_4}$	(iii)	Haloform reaction
(D)	$R-CH_2COOH \xrightarrow[\text{(ii) } H_2O]{\text{(i) } X_2\text{/Red P}}$	(iv)	Esterification

Choose the correct answer from the options given below.

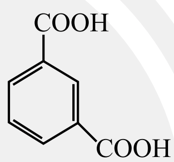
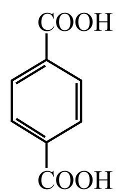
a. A-iii B-ii C-i D-iv

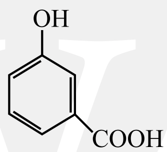
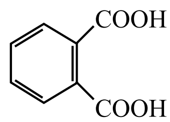
b. A-i B-iv C-iii D-ii

c. A-ii B-iii C-iv D-i

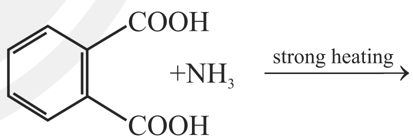
d. A-iv B-i C-ii D-iii

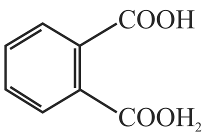
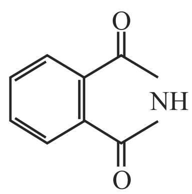
38. Which of the following acid will form an (i) Anhydride on heating and (ii) Acid imide on strong heating with ammonia? *(2020-Covid)*

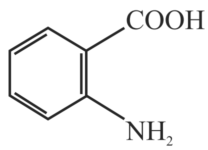
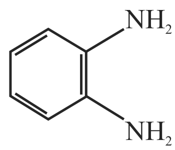
a.  b. 

c.  d. 

39. The major product of the following reaction is: *(2019)*



a.  b. 

c.  d. 

40. Carboxylic acids have higher boiling points than aldehydes, ketones and even alcohols of comparable molecular mass. It is due to their: *(2018)*

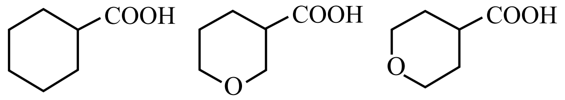
a. Formation of intramolecular H-bonding

b. Formation of carboxylate ion

c. Formation of intermolecular H-bonding

d. More extensive association of carboxylic acid via van der Waals force of attraction

41. The correct order of strengths of the carboxylic Acids is: *(2016 - II)*



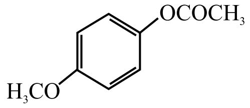
I II III

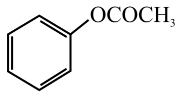
a. III > II > I
b. II > I > III
c. I > II > III
d. II > III > I

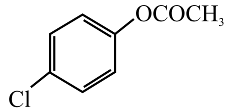
42. The oxidation of benzene by V_2O_5 in the presence of air produces: *(2015 Re)*

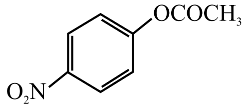
a. Maleic anhydride
b. Benzoic acid
c. Benzaldehyde
d. Benzoic anhydride

43. Which one of the following esters gets hydrolysed most easily under alkaline conditions? *(2015 Re)*

a. 

b. 

c. 

d. 

Answer Key

1. (a)	**2.** (d)	**3.** (d)	**4.** (a)	**5.** (d)	**6.** (b)	**7.** (d)	**8.** (d)	**9.** (a)	**10.** (c)
11. (b)	**12.** (d)	**13.** (b)	**14.** (d)	**15.** (a)	**16.** (b)	**17.** (b)	**18.** (a)	**19.** (c)	**20.** (c)
21. (c)	**22.** (d)	**23.** (b)	**24.** (b)	**25.** (c)	**26.** (b)	**27.** (c)	**28.** (b)	**29.** (d)	**30.** (b)
31. (a)	**32.** (b)	**33.** (b)	**34.** (c)	**35.** (c)	**36.** (b)	**37.** (c)	**38.** (d)	**39.** (b)	**40.** (c)
41. (d)	**42.** (a)	**43.** (d)							